CS-UY 4563 Final Project Report

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https://github.com/patrickxypluto/NYU-CS-UY-4564-Machine-Learning-Final-Project

1. Introduction

1.1 Dataset description

The dataset is a collection of Asian and Non-Asian face images sourced from Mendeley Data, containing 1,382 Asian images and 1,725 Non-Asian images, for a total of 3,107 samples. Each image is a 64×64 RGB photograph, yielding 12,288 features per sample. The dataset exhibits a mild class imbalance (44.5% Asian vs. 55.5% Non-Asian), which we account for when interpreting precision and recall metrics. The dataset can be accessed via the link provided in the reference section.

1.2 Machine-learning task

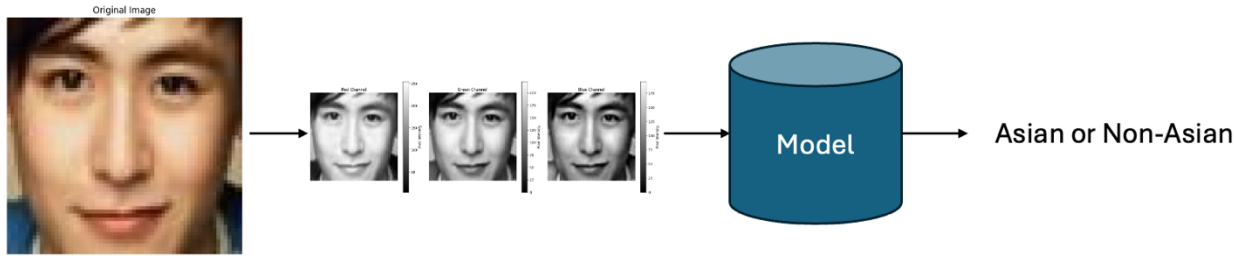


Figure 1. Classification of Asians and Non-Asians

Our goal is to build a robust binary classifier that distinguishes Asian from Non-Asian faces based solely on pixel information. This is an interesting and challenging task because human facial features vary continuously across ethnicities, making clean linear separation unlikely. It also serves as a practical testbed for comparing how well classical models like logistic regression handle high-dimensional image data versus more expressive architectures like neural networks and CNNs.

We apply three families of models — logistic regression, fully connected neural networks, and convolutional neural networks — each tested under multiple feature transformations and regularization settings. Model performance is evaluated using accuracy, precision, recall, and specificity on a held-out validation set, with the dataset split into 70% training, 15% validation, and 15% test.

2. Exploratory & Unsupervised Analysis

Before training any model, we explore the dataset to understand its structure and distribution.

2.1 Mean Face & Pixel Difference

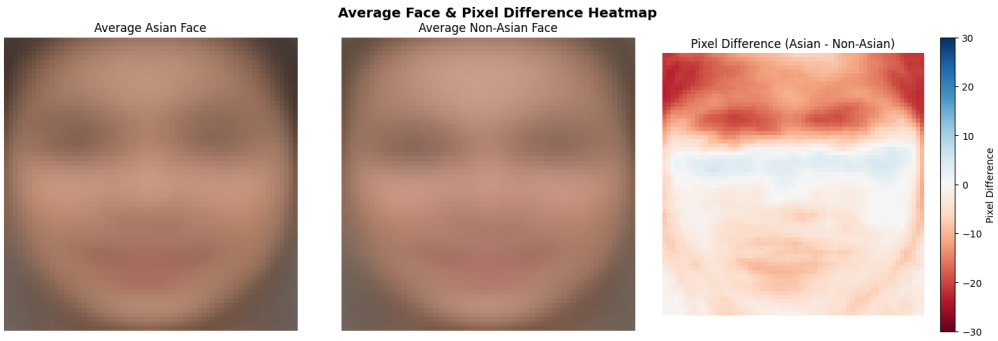


Figure 2.1 Average Face & Pixel Heatmap

To gain an intuitive understanding of class-level differences, we compute the per-pixel average across all images in each class and visualize the result alongside a pixel difference heatmap (Fig 2.1). Warmer colors in the heatmap indicate regions where Asian faces tend to be brighter on average, while cooler colors indicate the opposite.

The most intense colors appear around the eye region and the upper-left and upper-right corners of the head, suggesting that differences in eye appearance and hair width or density are the primary visual characteristics distinguishing the two groups at the pixel level.

2.2 Data Preprocessing

Before training any model, we normalize all pixel values by dividing by 255, scaling each value from [0, 255] to [0, 1]. This ensures that no single pixel dominates due to scale alone and stabilizes gradient-based optimization. We then split the dataset into 70% training, 15% validation, and 15% test sets. All preprocessing — including PCA fitting — is performed exclusively on the training set and then applied to the validation and test sets to prevent data leakage.

2.3. Principle Component Analysis

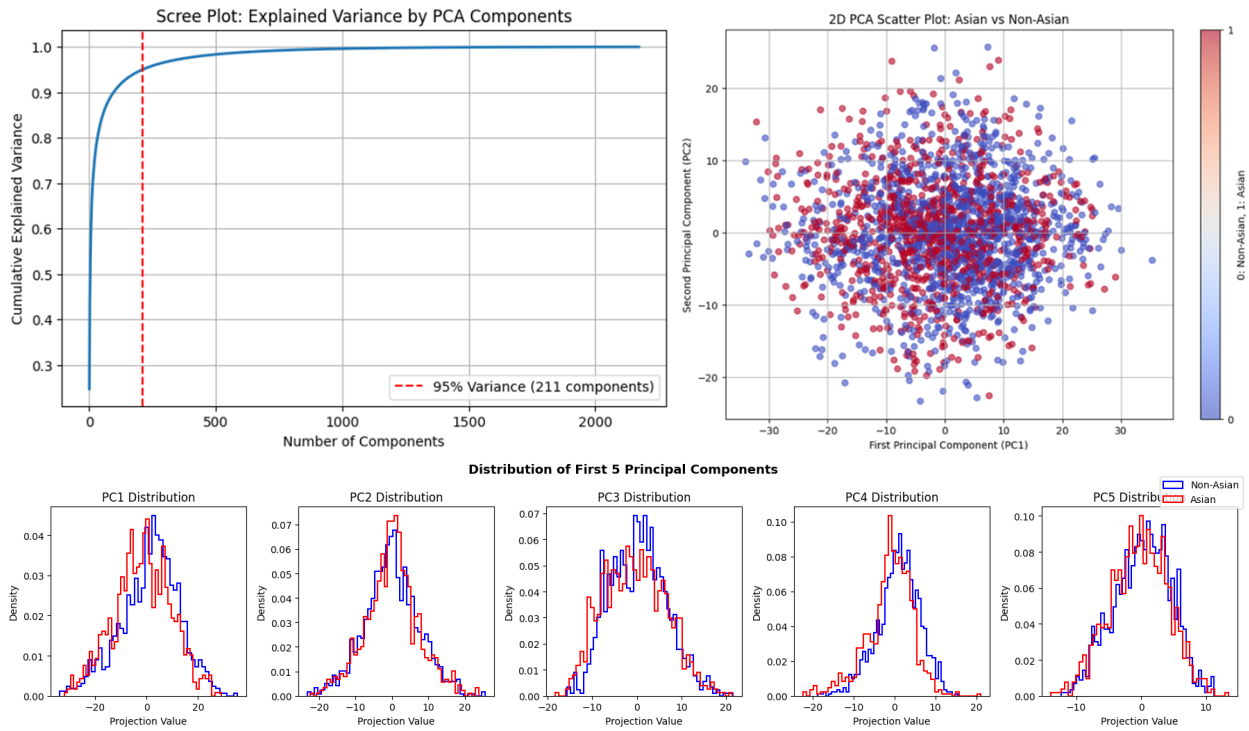


Figure 2.2 PCA

Since each raw 64×64 RGB image contains 12,288 features, the data is extremely high-dimensional. We apply PCA to reduce this dimensionality while retaining as much variance as possible. As shown in the scree plot (Fig 2.2 upleft), 211 principal components are sufficient to explain 95% of the total variance, confirming that the data can be significantly compressed with minimal information loss.

However, the 2D PCA scatter plot (Fig 2.2 upright) reveals substantial overlap between Asian (red) and Non-Asian (blue) samples along the first two principal components, indicating that linear projections alone are insufficient to separate the two classes. This motivates the use of more expressive nonlinear models.

The distributions of the first five principal components (Fig 2.2 below) further reinforce this challenge. While most components show heavily overlapping distributions between the two classes, slight shifts in peak density — particularly in PC1 and PC4 — hint at underlying structure that a sufficiently complex classifier could exploit.

3. Logistic Regression

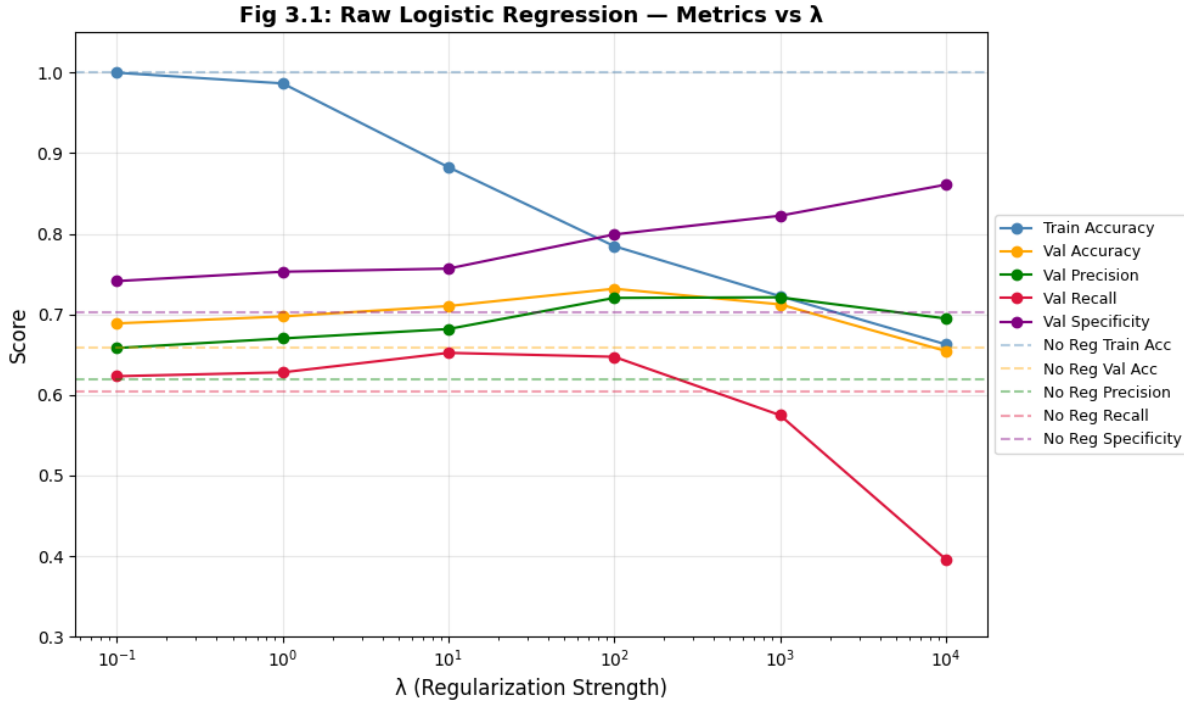
Logistic regression is our baseline supervised model. Despite its simplicity, it provides a strong reference point for evaluating how much more expressive models gain over a linear decision boundary. In this section, we test logistic regression under three feature transformations plus the untransformed raw input, sweeping L2 regularization strength (λ) across six values for each.

3.1 Untransformed (Raw Pixels)

Table 3.1: Raw Logistic Regression — All Metrics by λ

	λ	Train Acc	Val Acc	Precision	Recall	Specificity
0	0 (No Reg)	1.0000	0.6588	0.6188	0.6039	0.7027
1	0.1	1.0000	0.6888	0.6582	0.6232	0.7413
2	1	0.9867	0.6974	0.6701	0.6280	0.7529
3	10	0.8827	0.7103	0.6818	0.6522	0.7568
4	100	0.7847	0.7318	0.7204	0.6473	0.7992
5	1000	0.7226	0.7124	0.7212	0.5749	0.8224
6	10000	0.6628	0.6545	0.6949	0.3961	0.8610

Fig 3.1: Raw Logistic Regression — Metrics vs λ



We first train logistic regression directly on the normalized flattened 64×64 RGB images, giving each sample 12,288 features. This extremely high-dimensional input is expected to cause severe overfitting, as the model has far more parameters than can be reliably estimated from ~2,100 training samples.

Without regularization, the model achieves perfect training accuracy (1.0000) but only 0.6588 validation accuracy — a gap of over 34 percentage points. This massive train-val discrepancy is a textbook sign of overfitting: the model has memorized the training data rather than learning generalizable patterns.

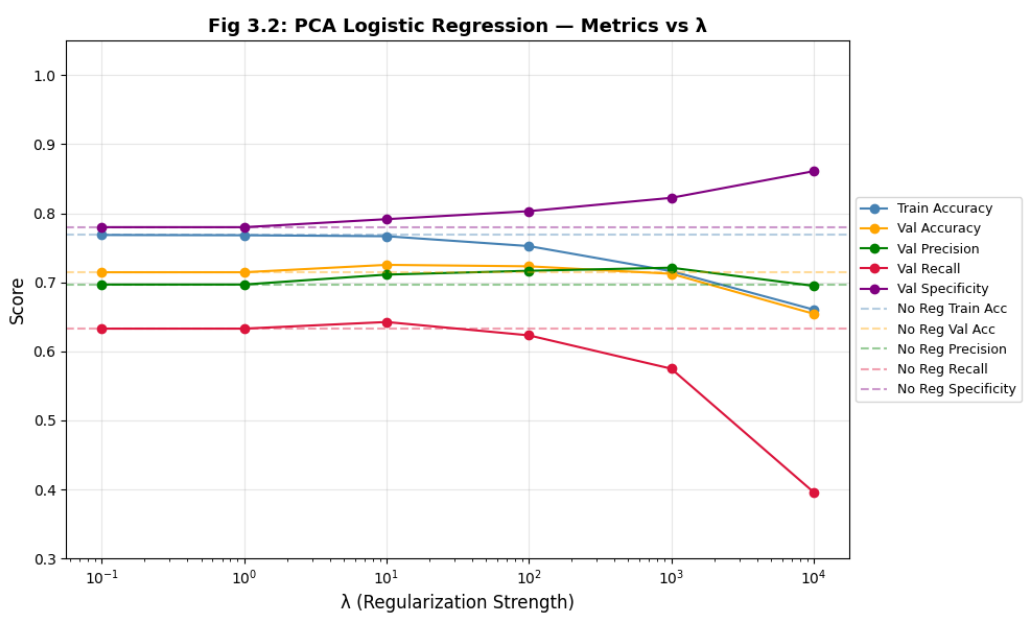
With L2 regularization, we sweep λ from 0.1 to 10,000 to progressively constrain the model. As shown in Fig 3.1 and Table 3.1, increasing λ steadily reduces training accuracy while improving validation accuracy up to a point, confirming that regularization is successfully reducing variance. The sweet spot is λ = 100, which achieves the best validation accuracy of 0.7318 with a balanced precision of 0.7204 and recall of 0.6473.

However, beyond λ = 100, over-regularization sets in. At λ = 10,000, recall collapses to 0.3961 while specificity rises to 0.8610 — the model becomes overly conservative, predicting Non-Asian for almost everything. This precision-recall tradeoff reveals that excessive regularization introduces too much bias, pushing the model toward the majority class.

3.2 Transformation 1: PCA (95% Variance)

Table 3.2: PCA Logistic Regression — All Metrics by λ

	λ	Train Acc	Val Acc	Precision	Recall	Specificity
0	0 (No Reg)	0.7686	0.7146	0.6968	0.6329	0.7799
1	0.1	0.7686	0.7146	0.6968	0.6329	0.7799
2	1	0.7682	0.7146	0.6968	0.6329	0.7799
3	10	0.7668	0.7253	0.7112	0.6425	0.7915
4	100	0.7525	0.7232	0.7167	0.6232	0.8031
5	1000	0.7162	0.7124	0.7212	0.5749	0.8224
6	10000	0.6605	0.6545	0.6949	0.3961	0.8610



The raw pixel space suffers from extremely high dimensionality relative to the number of training samples, making overfitting nearly unavoidable for a linear model. As a first transformation, we apply PCA and retain the top 211 components that collectively explain 95% of the total variance, reducing the feature space from 12,288 dimensions down to 211. This dimensionality reduction serves two purposes: it removes noisy, low-variance directions that likely hurt generalization, and it significantly speeds up training.

Without regularization, the PCA logistic model already achieves a training accuracy of 0.7686 and validation accuracy of 0.7146 — a gap of only ~5 percentage points compared to the 34-point gap seen in the raw model. This dramatic improvement confirms that PCA effectively reduces variance by discarding dimensions that the model would otherwise overfit to. The tradeoff, however, is a slight increase in bias: by compressing the feature space, some discriminative information is inevitably lost.

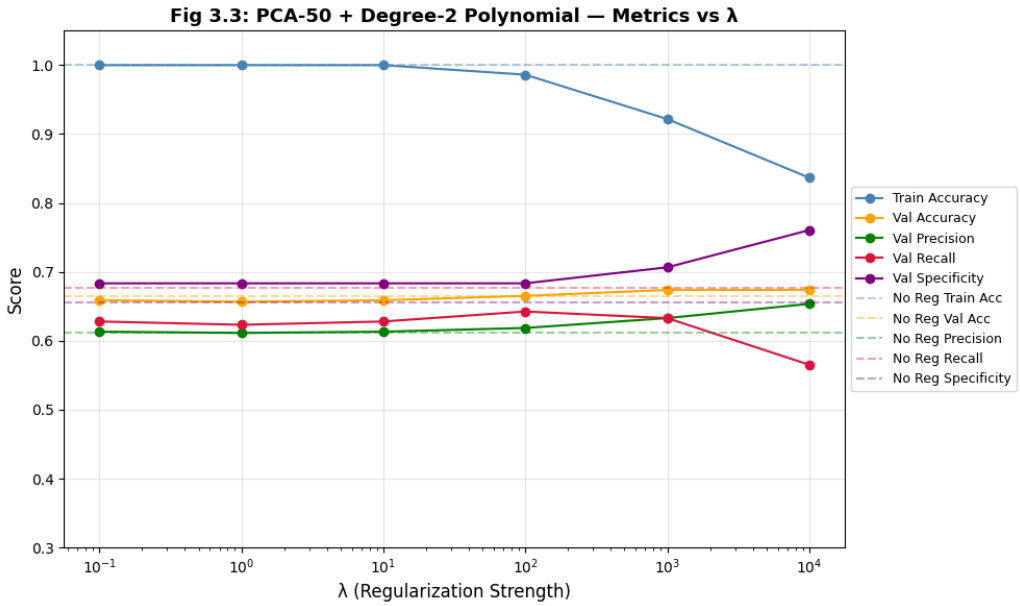
With L2 regularization, the model improves modestly. As shown in Fig 3.2 and Table 3.2, the best performance is achieved at $\lambda = 10$, yielding a validation accuracy of 0.7253, precision of 0.7112, and recall of 0.6425. Interestingly, the metrics are relatively stable across $\lambda = 0.1$ to $\lambda = 10$, suggesting the PCA transformation has already done much of the regularization work by reducing the hypothesis space. Beyond $\lambda = 1000$, the same over-regularization pattern emerges as in the raw model — recall collapses while specificity rises sharply, indicating the model again biases toward predicting the majority class.

Comparing with 3.1, a striking finding emerges: the raw logistic model at $\lambda = 100$ (Val Acc: 0.7318) actually outperforms the best PCA logistic model (Val Acc: 0.7253) across all metrics. This suggests that while PCA successfully eliminates noise and reduces variance, it also discards some genuinely discriminative pixel-level information. L2 regularization on the raw features proves to be a more effective strategy here — it suppresses noise without throwing away signal entirely. This is a clear illustration of the bias-variance tradeoff: PCA lowers variance but raises bias, while L2 on raw pixels manages variance more surgically.

3.3 Transformation 2: PCA-50 + Degree-2 Polynomial

Table 3.3: PCA-50 + Degree-2 Polynomial Logistic — All Metrics by λ

	λ	Train Acc	Val Acc	Precision	Recall	Specificity
0	0 (No Reg)	1.0000	0.6652	0.6114	0.6763	0.6564
1	0.1	1.0000	0.6588	0.6132	0.6280	0.6834
2	1	1.0000	0.6567	0.6114	0.6232	0.6834
3	10	1.0000	0.6588	0.6132	0.6280	0.6834
4	100	0.9862	0.6652	0.6186	0.6425	0.6834
5	1000	0.9218	0.6738	0.6329	0.6329	0.7066
6	10000	0.8367	0.6738	0.6536	0.5652	0.7606



While PCA reduces noise effectively, it operates purely within a linear feature space. To explore whether capturing nonlinear interactions between principal components can improve classification, we apply a degree-2 polynomial expansion on top of PCA. Specifically, we first reduce the data to the top 50 principal components, then generate all degree-2 polynomial combinations, yielding 1,326 new features. The motivation is that facial distinctions between classes may not be linearly separable in PCA space, and quadratic interactions between components could capture more complex boundaries.

We chose 50 components rather than the full 211 because applying a degree-2 expansion to 211 components would produce over 22,000 features — far too costly and prone to overfitting given our training set size.

Without regularization, the model immediately overfits to a perfect training accuracy of 1.0000 but achieves only 0.6652 validation accuracy. This is actually worse than both the raw and PCA models without regularization, which is alarming given that we added more features. The

polynomial expansion has dramatically increased model complexity, giving logistic regression far more degrees of freedom than it can reliably use with ~2,100 training samples.

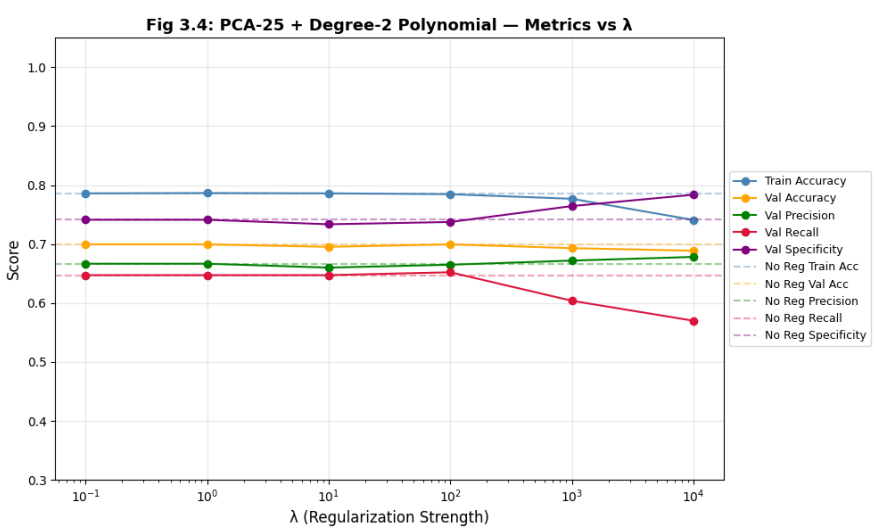
With L2 regularization, as shown in Fig 3.3 and Table 3.3, the results remain disappointing. Even at the best regularization value of λ = 1000, validation accuracy only reaches 0.6738 — well below both previous transformations. The train-val gap remains large throughout, and no amount of regularization manages to close it meaningfully. The recall in particular stays low, indicating the model struggles to correctly identify Asian faces across all settings.

The root cause is a classic high variance problem: the 1,326 polynomial features introduce a massive number of spurious interactions that are not genuinely discriminative for this task. While degree-2 polynomial expansion can be powerful for low-dimensional tabular data, it is poorly suited here because the top 50 PCA components already represent compressed global image statistics — their pairwise products do not correspond to meaningful facial features. Regularization can partially suppress this noise but cannot recover the information lost by discarding 161 PCA components and cannot eliminate the curse of dimensionality introduced by the polynomial expansion.

3.4 Transformation 3: PCA-25 + Degree-2 Polynomial

Table 3.4: PCA-25 + Degree-2 Polynomial Logistic — All Metrics by λ

	λ	Train Acc	Val Acc	Precision	Recall	Specificity
0	0 (No Reg)	0.7861	0.6996	0.6667	0.6473	0.7413
1	0.1	0.7861	0.6996	0.6667	0.6473	0.7413
2	1	0.7866	0.6996	0.6667	0.6473	0.7413
3	10	0.7861	0.6953	0.6601	0.6473	0.7336
4	100	0.7847	0.6996	0.6650	0.6522	0.7375
5	1000	0.7769	0.6931	0.6720	0.6039	0.7645
6	10000	0.7410	0.6888	0.6782	0.5700	0.7838



The poor performance of the PCA-50 polynomial model raised a natural question: is the problem the polynomial expansion itself, or the excessive number of features it generates? To investigate, we reduce the PCA components further to the top 25 before applying the degree-2 polynomial expansion, yielding 351 new features — roughly one quarter of the 1,326 features in the previous transformation. By keeping the feature count more manageable, we hope to retain some nonlinear expressiveness while reducing the risk of overfitting.

Without regularization, the model achieves a training accuracy of 0.7861 and validation accuracy of 0.6996 — a gap of ~9 percentage points. This is a significant improvement over the PCA-50 polynomial model, where the unregularized gap was over 33 points. Reducing the number of polynomial features has clearly helped control variance, confirming that the explosion of spurious interactions was indeed the main culprit in the previous transformation.

With L2 regularization, as shown in Fig 3.4 and Table 3.4, the metrics remain remarkably stable across all λ values from 0 to 100, with validation accuracy hovering around 0.6953–0.6996. This plateau suggests the model has already found a relatively stable solution in this compressed feature space, and additional regularization offers little further benefit. Beyond $\lambda = 1000$, the familiar pattern of over-regularization returns — recall drops while specificity climbs, indicating the model again biases toward predicting Non-Asian.

The best performance is achieved at $\lambda = 0$ or $\lambda = 1$ (Val Acc: 0.6996, Precision: 0.6667, Recall: 0.6473, Specificity: 0.7413), which is still notably worse than the raw logistic model at $\lambda = 100$.

This outcome reveals an important insight: reducing from 50 to 25 PCA components before the polynomial expansion significantly reduces overfitting, but at the cost of discarding even more discriminative information. The top 25 components capture only a fraction of the variance retained by 50 or 211 components, meaning the nonlinear interactions being modeled are built on an increasingly impoverished representation of the original image. The polynomial expansion adds complexity on top of an already lossy compression, and the net effect is a model that neither generalizes well nor captures meaningful nonlinear structure. This confirms that degree-2 polynomial expansions on PCA-compressed image data are not an effective strategy for this task, regardless of how many components are retained.

3.5 Logistic Regression Summary

Table 3.5: Logistic Regression — Best Model Per Transformation

	Transformation	Best λ	Train Acc	Val Acc	Precision	Recall	Specificity
0	Raw Pixels	100	0.7847	0.7318	0.7204	0.6473	0.7992
1	PCA (211 components)	10	0.7668	0.7253	0.7112	0.6425	0.7915
2	PCA-50 + Poly-2	1000	0.9218	0.6738	0.6329	0.6329	0.7066
3	PCA-25 + Poly-2	0	0.7861	0.6996	0.6667	0.6473	0.7413

Across all four configurations, the best logistic regression model is the Raw Logistic at $\lambda = 100$, achieving a validation accuracy of 0.7318, precision of 0.7204, recall of 0.6473, and specificity of 0.7992. This result is counterintuitive at first glance — one might expect dimensionality

reduction or nonlinear feature expansion to help — but it reflects a consistent theme throughout this section: for this particular task, L2 regularization on raw pixels is a more effective variance-reduction strategy than aggressive feature preprocessing.

PCA reduces the train-val gap dramatically but discards discriminative information, raising bias. Polynomial expansions on PCA components introduce too many spurious interactions relative to the training set size, causing severe overfitting that regularization cannot fully remedy. The raw pixel representation, despite its high dimensionality, preserves all available signal, and a well-tuned L2 penalty is sufficient to prevent the model from memorizing noise.

That said, all logistic regression models are fundamentally limited by their linear decision boundary. Given the heavy overlap observed in the PCA scatter plot (Fig 2.2), it is unlikely that any linear classifier can achieve strong performance on this task regardless of feature transformation or regularization. This motivates the use of neural networks and CNNs in the following sections.

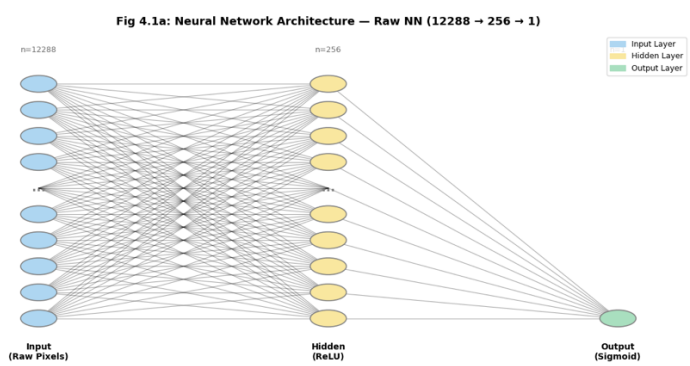
4. Neural network

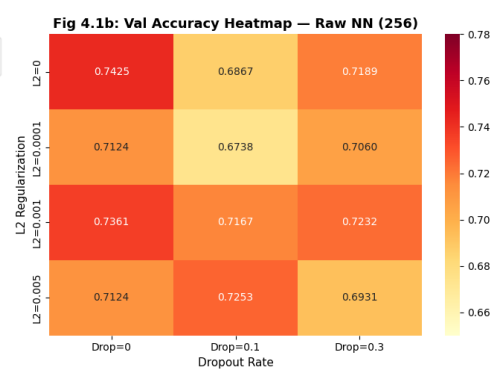
While logistic regression is limited to linear decision boundaries, fully connected neural networks can learn nonlinear representations through multiple layers of transformations. In this section, we train four neural network architectures of increasing or varying complexity, treating each distinct architecture as a feature transformation of the input. All models are trained using the Adam optimizer with binary cross-entropy loss, ReLU hidden activations, sigmoid output activation, 50 epochs, and a batch size of 32. For each architecture, we sweep over four L2 weight decay values (0, 0.0001, 0.001, 0.005) and three dropout rates (0, 0.1, 0.3), giving 12 regularization combinations per architecture.

4.1 Single Hidden Layer: Raw NN (12288→256→1)

Table 4.1: Raw NN (12288→256→1) — All Regularization Combinations

	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	0.797100	0.742500	0.748600	0.632900	0.830100
1	0.0000	0.1	0.759000	0.686700	0.612500	0.801900	0.594600
2	0.0000	0.3	0.753400	0.718900	0.704300	0.632900	0.787600
3	0.0001	0.0	0.762600	0.712400	0.732500	0.555600	0.837800
4	0.0001	0.1	0.713000	0.673800	0.789500	0.362300	0.922800
5	0.0001	0.3	0.735100	0.706000	0.757400	0.497600	0.872600
6	0.0010	0.0	0.787900	0.736100	0.772700	0.574900	0.864900
7	0.0010	0.1	0.752100	0.716700	0.672800	0.705300	0.725900
8	0.0010	0.3	0.745200	0.723200	0.705300	0.647300	0.783800
9	0.0050	0.0	0.759900	0.712400	0.732500	0.555600	0.837800
10	0.0050	0.1	0.736000	0.725300	0.788300	0.521700	0.888000
11	0.0050	0.3	0.681700	0.693100	0.675800	0.594200	0.772200





As our baseline neural network, we use a single hidden layer with 256 units fed directly by the normalized raw pixel input of 12,288 features. This architecture is intentionally simple — we want to establish how much a minimal nonlinear model improves over logistic regression before adding complexity.

Without any regularization, the model achieves a training accuracy of 0.7971 and validation accuracy of 0.7425. Compared to the best logistic regression model (Val Acc: 0.7318), this is already an improvement, suggesting that even a single nonlinear layer captures useful structure that a linear boundary cannot. The train-val gap of ~5 points is moderate, indicating mild overfitting.

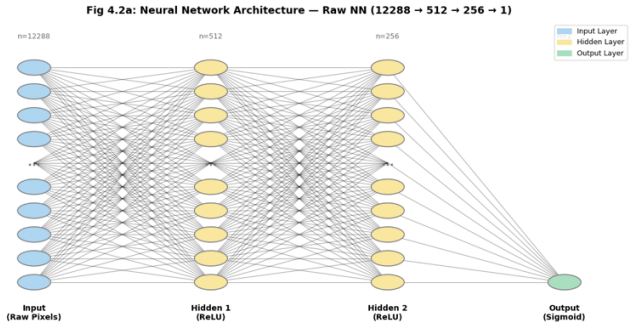
With regularization, results are mixed. As shown in Table 4.1, adding dropout alone (L2=0, Dropout=0.1) actually hurts performance, dropping validation accuracy to 0.6867 while pushing recall up to 0.8019 — the model becomes overly sensitive, predicting Asian too aggressively. This suggests dropout at this rate is too destructive for such a wide single layer. Combining L2=0.001 with no dropout achieves a strong validation accuracy of 0.7361 with a well-balanced precision of 0.7727, but at the cost of lower recall (0.5749), meaning it misses more Asian faces. The best overall configuration remains L2=0, Dropout=0, confirming that the baseline architecture is already close to its capacity limit and regularization tends to hurt more than help here.

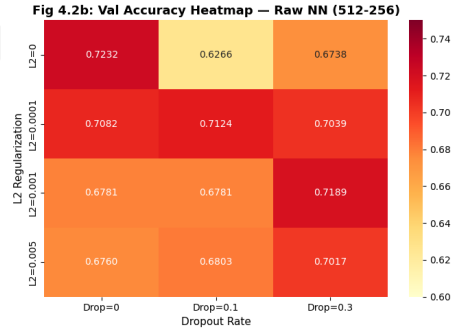
The relatively modest train-val gap without regularization suggests this architecture sits in a reasonable bias-variance balance for raw pixel input — complex enough to learn useful patterns, but not so large as to massively overfit.

4.2 Two Wide Hidden Layers: Raw NN (12288→512→256→1)

Table 4.2: Raw NN (12288→512→256→1) — All Regularization Combinations

	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	0.803100	0.723200	0.691200	0.681200	0.756800
1	0.0000	0.1	0.676200	0.626600	0.548700	0.898600	0.409300
2	0.0000	0.3	0.714800	0.673800	0.795700	0.357500	0.926600
3	0.0001	0.0	0.818300	0.708200	0.649800	0.744000	0.679500
4	0.0001	0.1	0.750700	0.712400	0.652700	0.753600	0.679500
5	0.0001	0.3	0.716200	0.703900	0.656100	0.700500	0.706600
6	0.0010	0.0	0.721300	0.678100	0.787900	0.376800	0.918900
7	0.0010	0.1	0.746600	0.678100	0.602900	0.806800	0.575300
8	0.0010	0.3	0.759000	0.718900	0.681000	0.690800	0.741300
9	0.0050	0.0	0.723100	0.676000	0.601400	0.801900	0.575300
10	0.0050	0.1	0.703800	0.680300	0.754400	0.415500	0.891900
11	0.0050	0.3	0.723100	0.701700	0.675300	0.632900	0.756800





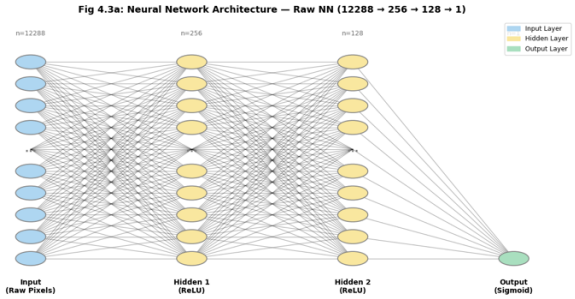
Having established a single hidden layer baseline, we now increase model capacity by adding a second hidden layer and widening the first layer from 256 to 512 units. The motivation is that a deeper, wider network should be able to learn more complex hierarchical representations of the raw pixel input, potentially capturing higher-order facial patterns that a single layer cannot.

Without any regularization, the model achieves a training accuracy of 0.8031 but validation accuracy drops to 0.7232 compared to 0.7425 in the shallower model. This is a clear sign that the added capacity is causing increased overfitting — the wider, deeper network has more parameters to memorize training examples rather than generalizing. The train-val gap has widened relative to 4.1, confirming our suspicion.

With regularization, the results remain inconsistent. As shown in Table 4.2, no regularization combination manages to surpass the unregularized single-layer model. Some configurations produce extreme behavior — for example, L2=0, Dropout=0.1 yields a recall of 0.8986 but precision of only 0.5487, meaning the model is predicting Asian for nearly everything. Conversely, L2=0, Dropout=0.3 collapses recall to 0.3575 while pushing specificity to 0.9266 — the opposite extreme. This wild swings in precision and recall across regularization settings indicate the model is highly unstable, a symptom of a network that is too large for the available data.

The heatmap (Fig 4.2b) confirms that no regularization combination consistently improves validation accuracy beyond 0.7232. The best result is L2=0, Dropout=0 (Val Acc: 0.7232), which is still worse than the simpler 4.1 architecture. This strongly suggests that for raw pixel input with ~2,100 training samples, wider and deeper is not better — the additional parameters simply provide more opportunities to overfit without learning more generalizable features.

4.3 Narrower Two Wide Hidden Layers: Raw NN (12288→256→128→1)



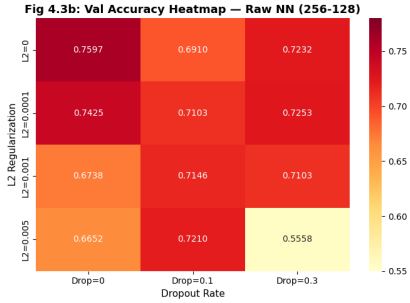


Table 4.3: Raw NN (12288→256→128→1) — All Regularization Combinations

	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	0.809100	0.759700	0.765400	0.661800	0.837800
1	0.0000	0.1	0.777400	0.691000	0.622600	0.772900	0.625500
2	0.0000	0.3	0.749800	0.723200	0.721600	0.613500	0.810800
3	0.0001	0.0	0.825200	0.742500	0.695100	0.748800	0.737500
4	0.0001	0.1	0.792500	0.710300	0.662200	0.710100	0.710400
5	0.0001	0.3	0.747000	0.725300	0.713500	0.637700	0.795400
6	0.0010	0.0	0.718500	0.673800	0.777800	0.372000	0.915100
7	0.0010	0.1	0.730500	0.714600	0.764300	0.516900	0.872600
8	0.0010	0.3	0.722200	0.710300	0.736800	0.541100	0.845600
9	0.0050	0.0	0.708400	0.665200	0.587000	0.830900	0.532800
10	0.0050	0.1	0.750200	0.721000	0.710400	0.628000	0.795400
11	0.0050	0.3	0.555200	0.555800	0.000000	0.000000	1.000000

The previous architecture's instability and overfitting suggest that 512 units in the first hidden layer is excessive for this task. We therefore narrow the network, keeping two hidden layers but reducing their sizes to 256 and 128 units respectively. The intuition is that a more compact two-layer architecture can still learn hierarchical nonlinear features while having fewer parameters to overfit with, striking a better balance between model capacity and generalization.

Without any regularization, the model achieves a training accuracy of 0.8091 and validation accuracy of 0.7597 — the best unregularized performance seen so far across all neural network architectures. The train-val gap of ~5 points is comparable to the single-layer model but with a higher ceiling, suggesting that the two-layer narrower architecture successfully learns more useful representations without immediately collapsing into memorization.

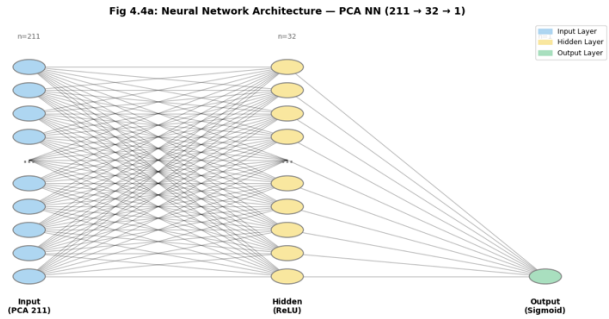
With regularization, as shown in Table 4.3, performance is more nuanced. The best overall result is L2=0, Dropout=0 (Val Acc: 0.7597, Precision: 0.7654, Recall: 0.6618, Specificity: 0.8378), confirming once again that this architecture is already near its sweet spot without explicit regularization. Adding small amounts of L2 (0.0001) with no dropout yields a slightly higher recall of 0.7488 at the cost of lower precision (0.6951), reflecting a shift in the decision threshold rather than a genuine improvement in generalization. Stronger regularization (L2=0.001, Dropout=0) causes recall to collapse to 0.3720 while specificity jumps to 0.9151 — the model becomes overly conservative again.

One particularly notable failure is L2=0.005, Dropout=0.3, which produces a training accuracy of 0.5552, validation accuracy of 0.5558, precision of 0.0000, recall of 0.0000, and specificity of 1.0000. This means the model has completely degenerated — it predicts Non-Asian for every single sample. This is a textbook case of extreme over-regularization: the combined effect of

strong L2 weight decay and aggressive dropout has suppressed all learned weights to near zero, causing the model to default to the majority class prediction at all times.

The heatmap (Fig 4.3b) clearly shows that the top-left region (low L2, low dropout) consistently yields the best validation accuracy, with performance degrading as regularization strength increases in either dimension. This architecture represents the best fully connected network on raw pixels found so far, confirming that moderate depth with controlled width outperforms both shallower and wider alternatives for this dataset.

4.4 PCA Input: PCA NN (211→32→1)



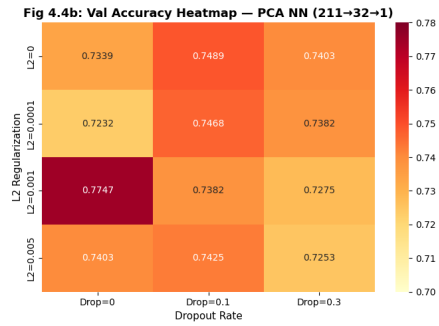


Table 4.4: PCA NN (211→32→1) — All Regularization Combinations

	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	1.000000	0.733900	0.700500	0.700500	0.760600
1	0.0000	0.1	0.999500	0.748900	0.722800	0.705300	0.783800
2	0.0000	0.3	0.996300	0.740300	0.724000	0.671500	0.795400
3	0.0001	0.0	1.000000	0.723200	0.693100	0.676300	0.760600
4	0.0001	0.1	1.000000	0.746800	0.740500	0.661800	0.814700
5	0.0001	0.3	0.997200	0.738200	0.715700	0.681200	0.783800
6	0.0010	0.0	1.000000	0.774700	0.750000	0.739100	0.803100
7	0.0010	0.1	0.999500	0.738200	0.705300	0.705300	0.764500
8	0.0010	0.3	0.997200	0.727500	0.712800	0.647300	0.791500
9	0.0050	0.0	1.000000	0.740300	0.710800	0.700500	0.772200
10	0.0050	0.1	0.996800	0.742500	0.723100	0.681200	0.791500
11	0.0050	0.3	0.982100	0.725300	0.711200	0.642500	0.791500

Having explored three architectures on raw pixel input, we now switch the input representation to PCA-compressed features. Instead of feeding 12,288 raw pixel values, we use the top 211 principal components that explain 95% of the total variance, and pair them with a compact single hidden layer of 32 units. The motivation is twofold: first, PCA removes low-variance noise directions that may confuse the network; second, a smaller hidden layer is more appropriate for a

211-dimensional input, avoiding the excessive parameter counts seen in the raw pixel architectures.

Without any regularization, the model achieves a perfect training accuracy of 1.0000 but a validation accuracy of 0.7339 — a gap of over 26 points. This is surprising given that PCA logistic regression showed almost no overfitting without regularization. The explanation lies in the nature of neural networks: even a small hidden layer of 32 units introduces enough nonlinear capacity to memorize the compressed PCA representation of the training set, whereas logistic regression is fundamentally constrained to a linear boundary. The PCA input has reduced the noise but the nonlinearity of the network compensates by overfitting more aggressively.

With regularization, as shown in Table 4.4, the results improve substantially and consistently. Unlike the raw pixel architectures where regularization often caused erratic swings in precision and recall, here the metrics are much more stable across regularization combinations. This stability reflects the cleaner, lower-dimensional input space — the network is working with more structured features and regularization can therefore operate more predictably.

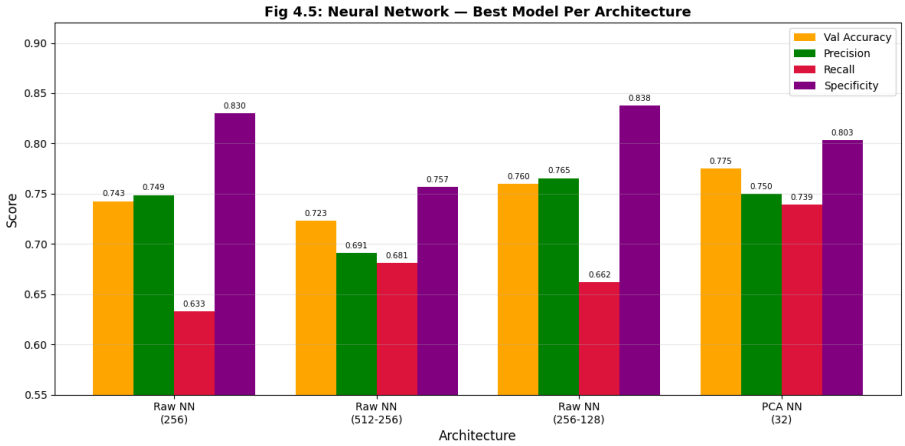
The best result is achieved at L2=0.001, Dropout=0 (Train Acc: 1.0000, Val Acc: 0.7747, Precision: 0.7500, Recall: 0.7391, Specificity: 0.8031). This is remarkable for two reasons. First, it is the best validation accuracy achieved by any neural network architecture so far. Second, and more importantly, it achieves the most balanced set of metrics across all configurations — precision, recall, and specificity are all relatively high simultaneously, suggesting the model is making genuinely good predictions rather than simply biasing toward one class. The recall of 0.7391 in particular is significantly higher than most raw pixel models, meaning it correctly identifies Asian faces at a much higher rate.Interestingly, even though training accuracy remains at 1.0000, the validation accuracy of 0.7747 represents strong generalization given the task difficulty. This suggests that L2 regularization is successfully constraining the weight magnitudes without destroying the useful nonlinear patterns learned from the PCA features. The combination of a compressed, denoised input representation with moderate L2 regularization proves to be the most effective strategy among all fully connected architectures tested.The heatmap (Fig 4.4b) clearly illustrates that the optimal regularization region is around L2=0.001 with low or no dropout, with performance degrading at higher L2 values as the model becomes increasingly underfitted. Dropout alone without L2 also helps modestly, with L2=0, Dropout=0.1 achieving a validation accuracy of 0.7489.

4.5 Neural Network Summary

Across all four architectures, the best neural network model is PCA NN (211 → 32 → 1) with L2=0.001, Dropout=0, achieving a validation accuracy of 0.7747, precision of 0.7500, recall of 0.7391, and specificity of 0.8031. This result highlights two key lessons learned throughout this section.

Table 4.5: Neural Network — Best Model Per Architecture

	Architecture	Input	Best L2	Best Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	Raw NN (256)	Raw Pixels	0.0000	0.0	0.797100	0.742500	0.748600	0.632900	0.830100
1	Raw NN (512-256)	Raw Pixels	0.0000	0.0	0.803100	0.723200	0.691200	0.681200	0.756800
2	Raw NN (256-128)	Raw Pixels	0.0000	0.0	0.809100	0.759700	0.765400	0.661800	0.837800
3	PCA NN (32)	PCA (211)	0.0010	0.0	1.000000	0.774700	0.750000	0.739100	0.803100



First, input representation matters more than architecture size. The PCA-based network, despite being the smallest architecture tested, outperforms all three raw pixel networks. By feeding the network cleaner, lower-dimensional features, we give it a better foundation to learn from, and regularization can operate more predictably and effectively. Raw pixel networks suffer from the curse of dimensionality — 12,288 input features overwhelm the network's ability to generalize, regardless of how many hidden layers are added.

Second, more capacity does not guarantee better performance. The widest and deepest raw pixel network (512 → 256) actually performed worst among the raw architectures, while the moderate (256 → 128) network performed best. This reflects a fundamental principle: model capacity should be matched to the complexity of the task and the size of the available training data. With only ~2,100 training samples, overly large networks have too many degrees of freedom and regularization alone cannot fully compensate.

That said, all fully connected networks share a common limitation — they treat each pixel or PCA component as an independent input feature, ignoring the inherent spatial structure of images. Pixels that are physically adjacent in a face are not treated any differently from pixels that are far apart. This is precisely the limitation that convolutional neural networks are designed to overcome, motivating the next section.

5. Convolutional Neural Network

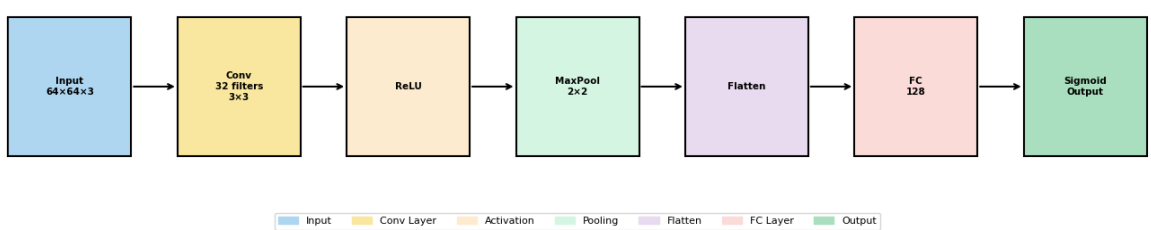
Unlike fully connected networks, CNNs exploit the spatial structure of images through local convolutional filters, making them naturally well-suited for image classification. Rather than treating each pixel independently, convolutional layers detect local patterns such as edges, textures, and shapes that are meaningful regardless of their position in the image. In this section, we test four CNN architectures of increasing complexity, all trained on normalized raw pixel data using the same optimizer, loss, and regularization grid as the neural network section.

5.1 Shallow CNN – CNN1

Table 5.1: CNN1 — All Regularization Combinations

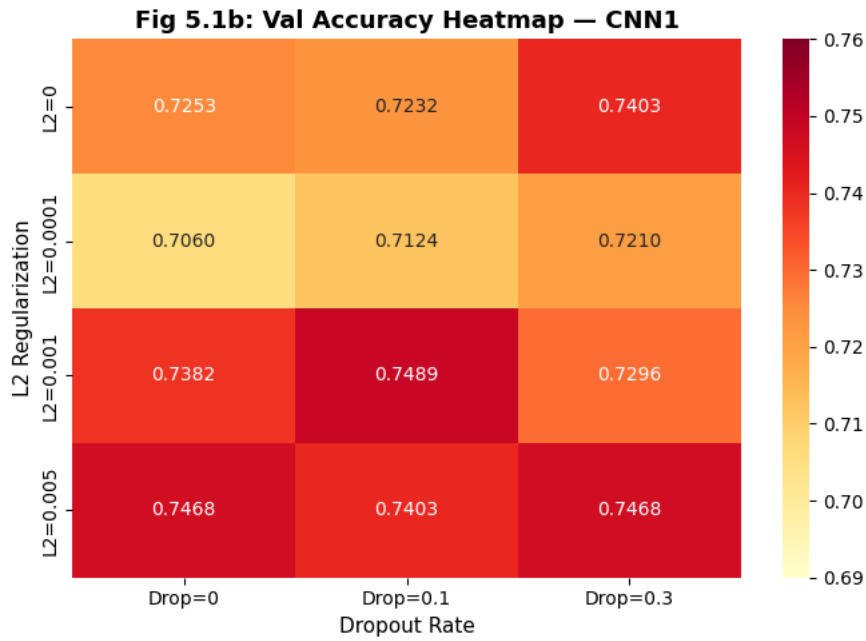
	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	1.000000	0.725300	0.687200	0.700500	0.745200
1	0.0000	0.1	0.998600	0.723200	0.712000	0.632900	0.795400
2	0.0000	0.3	1.000000	0.740300	0.710800	0.700500	0.772200
3	0.0001	0.0	1.000000	0.706000	0.675000	0.652200	0.749000
4	0.0001	0.1	0.977900	0.712400	0.721200	0.574900	0.822400
5	0.0001	0.3	1.000000	0.721000	0.680800	0.700500	0.737500
6	0.0010	0.0	0.989000	0.738200	0.742900	0.628000	0.826300
7	0.0010	0.1	0.992600	0.748900	0.706400	0.744000	0.752900
8	0.0010	0.3	0.980200	0.729600	0.840300	0.483100	0.926600
9	0.0050	0.0	0.918100	0.746800	0.757200	0.632900	0.837800
10	0.0050	0.1	0.943400	0.740300	0.693700	0.744000	0.737500
11	0.0050	0.3	0.886800	0.746800	0.790800	0.584500	0.876400

Fig 5.1a: CNN1 Architecture



Architecture: Conv(32, 3×3) → ReLU → MaxPool → Flatten → FC(128) → Sigmoid

As our CNN baseline, we use a single convolutional layer with 32 filters followed by a fully connected head. This minimal architecture tests whether even basic spatial feature extraction outperforms fully connected networks on raw pixels.

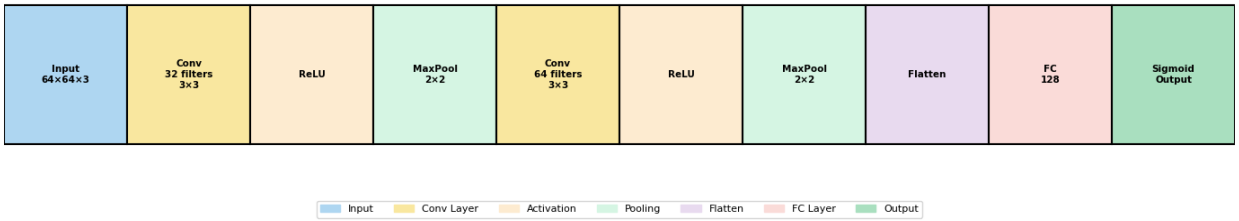


Without regularization, the model achieves a perfect training accuracy of 1.0000 but only 0.7253 validation accuracy, indicating overfitting. Despite this, it already matches or exceeds several fully connected architectures, confirming that convolutional filters extract more useful features from image data than raw pixel weights.

With regularization, as shown in Table 5.1, the best result is L2=0.001, Dropout=0.1 (Val Acc: 0.7489, Precision: 0.7064, Recall: 0.7440, Specificity: 0.7529) — a well-balanced set of metrics. The heatmap (Fig 5.1b) shows that moderate L2 combined with light dropout consistently improves over no regularization, suggesting the single conv layer benefits from both forms of constraint working together.

5.2 Two Convolutional Layers – CNN2

Fig 5.2a: CNN2 Architecture



Architecture: Conv(32) → ReLU → MaxPool → Conv(64) → ReLU → MaxPool → Flatten → FC(128) → Sigmoid

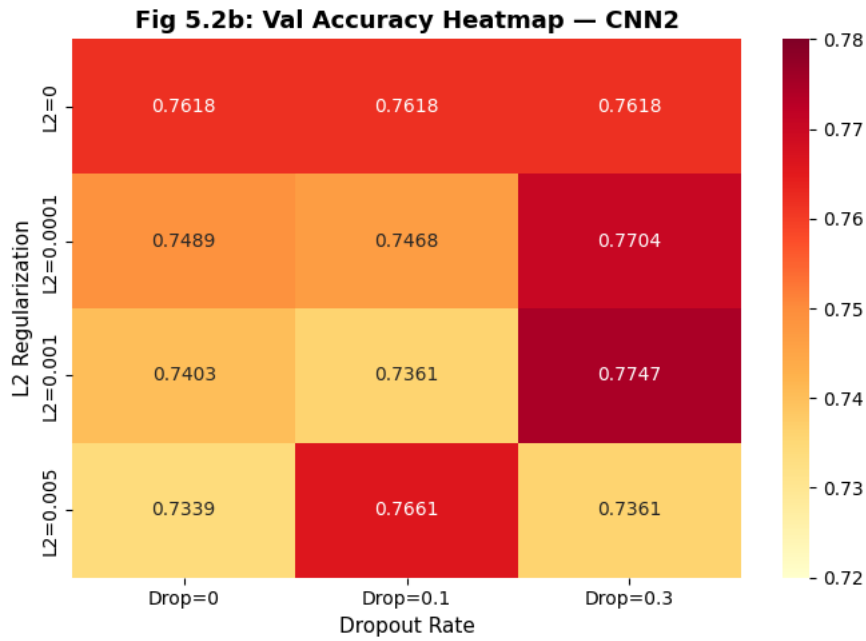


Table 5.2: CNN2 — All Regularization Combinations

	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	1.000000	0.761800	0.744900	0.705300	0.806900
1	0.0000	0.1	1.000000	0.761800	0.720200	0.758500	0.764500
2	0.0000	0.3	1.000000	0.761800	0.760900	0.676300	0.830100
3	0.0001	0.0	1.000000	0.748900	0.725000	0.700500	0.787600
4	0.0001	0.1	1.000000	0.746800	0.717100	0.710100	0.776100
5	0.0001	0.3	1.000000	0.770400	0.768800	0.690800	0.834000
6	0.0010	0.0	1.000000	0.740300	0.708700	0.705300	0.768300
7	0.0010	0.1	0.997700	0.736100	0.714300	0.676300	0.783800
8	0.0010	0.3	0.995400	0.774700	0.731800	0.777800	0.772200
9	0.0050	0.0	0.955800	0.733900	0.676600	0.768100	0.706600
10	0.0050	0.1	0.990300	0.766100	0.726900	0.758500	0.772200
11	0.0050	0.3	0.951700	0.736100	0.682600	0.758500	0.718100

We add a second convolutional layer with 64 filters to allow the network to learn hierarchical features — low-level edges in the first layer and higher-level patterns in the second. This is a natural progression from CNN1 and is expected to improve performance by capturing more abstract facial representations.

Without regularization, the model achieves a training accuracy of 1.0000 and validation accuracy of 0.7618 — a notable jump from CNN1's 0.7253. The second convolutional layer is clearly extracting more useful hierarchical features, validating the motivation for deeper convolutional architectures.

With regularization, as shown in Table 5.2, results are consistently strong across multiple configurations. Notably, L2=0.001, Dropout=0.3 achieves the best validation accuracy of 0.7747 with a well-balanced precision of 0.7318 and recall of 0.7778 — the highest recall seen in any CNN so far. The heatmap (Fig 5.2b) shows a more uniform performance landscape than CNN1, suggesting the two-layer architecture is more robust to different regularization choices. Unlike the fully connected networks, moderate dropout here helps rather than hurts, as the convolutional features are more redundant and can tolerate random deactivation better.

5.3 Three Convolutional Layers – CNN3

Fig 5.3a: CNN3 Architecture

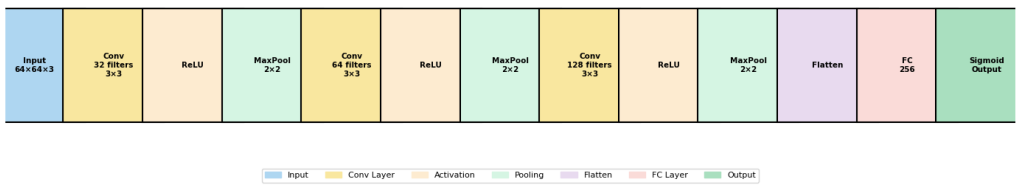
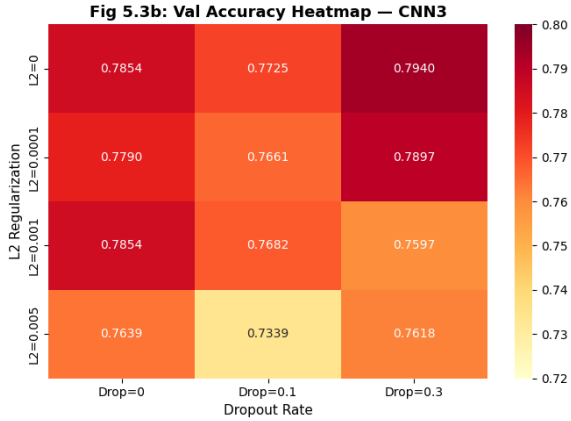


Table 5.3: CNN3 — All Regularization Combinations

	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	1.000000	0.785400	0.774400	0.729500	0.830100
1	0.0000	0.1	1.000000	0.772500	0.756300	0.719800	0.814700
2	0.0000	0.3	0.998600	0.794000	0.770700	0.763300	0.818500
3	0.0001	0.0	1.000000	0.779000	0.754900	0.744000	0.806900
4	0.0001	0.1	0.972400	0.766100	0.845100	0.579700	0.915100
5	0.0001	0.3	1.000000	0.789700	0.776600	0.739100	0.830100
6	0.0010	0.0	0.989000	0.785400	0.758500	0.758500	0.806900
7	0.0010	0.1	0.999500	0.768200	0.734600	0.748800	0.783800
8	0.0010	0.3	0.999100	0.759700	0.762400	0.666700	0.834000
9	0.0050	0.0	0.959100	0.763900	0.751300	0.700500	0.814700
10	0.0050	0.1	0.794400	0.733900	0.731800	0.632900	0.814700
11	0.0050	0.3	0.892800	0.761800	0.735300	0.724600	0.791500

Fig 5.3b: Val Accuracy Heatmap — CNN3



Architecture: Conv(32) → ReLU → MaxPool → Conv(64) → ReLU → MaxPool → Conv(128) → ReLU → MaxPool → Flatten → FC(256) → Sigmoid

We add a third convolutional layer with 128 filters and widen the fully connected head to 256 units. The motivation is to capture even more abstract, high-level facial representations through deeper feature hierarchies, at the cost of increased model complexity.

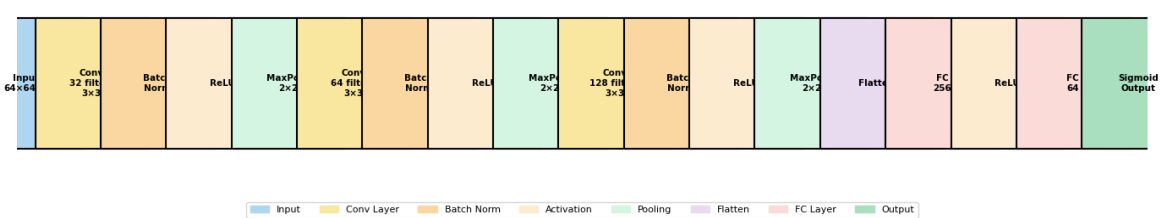
Without regularization, the model achieves a training accuracy of 1.0000 and validation accuracy of 0.7854 — a further improvement over CNN2's 0.7618. The deeper architecture is successfully extracting richer spatial features, and the larger FC head gives the network more capacity to combine them for classification.

With regularization, as shown in Table 5.3, performance remains strong and stable across many configurations. The best result is L2=0, Dropout=0.3 (Val Acc: 0.7940, Precision: 0.7707, Recall: 0.7633, Specificity: 0.8185) — notably, dropout alone without any L2 penalty achieves the peak here. This suggests that the third convolutional layer introduces enough redundancy in learned filters that dropout is the more natural regularizer, randomly deactivating features during training to prevent co-adaptation. The heatmap (Fig 5.3b) confirms that low-to-zero L2 combined with moderate dropout consistently performs well, while stronger L2 begins to hurt by over-constraining the convolutional filters.

Comparing with CNN2, the addition of a third convolutional layer provides a meaningful and consistent improvement across nearly all regularization settings, confirming that deeper feature hierarchies are genuinely beneficial for this task.

5.4 Three Convolutional Layers with Batch Normalization – CNN4

Fig 5.4a: CNN4 Architecture



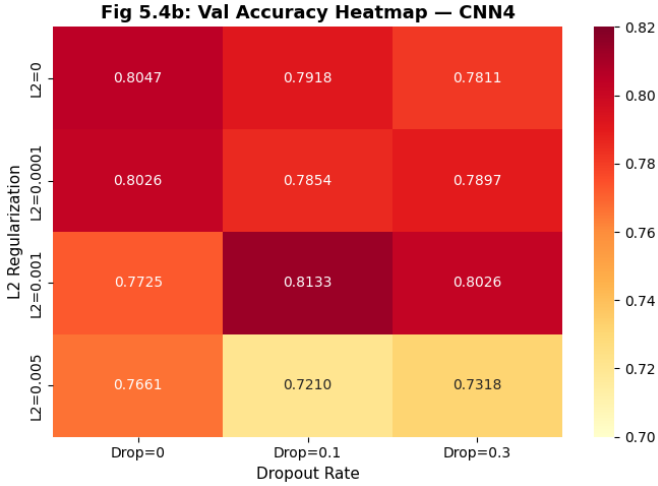
Architecture: Conv(32) → BN → ReLU → MaxPool → Conv(64) → BN → ReLU → MaxPool → Conv(128) → BN → ReLU → MaxPool → Flatten → FC(256) → ReLU → FC(64) → Sigmoid

Building on CNN3, we introduce batch normalization after each convolutional layer and deepen the fully connected head to two layers (256 → 64). Batch normalization normalizes activations within each mini-batch, stabilizing training and reducing internal covariate shift. The deeper FC head gives the network more capacity to combine learned spatial features for the final classification decision.

Table 5.4: CNN4 — All Regularization Combinations

	L2	Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	0.0000	0.0	1.000000	0.804700	0.781600	0.777800	0.826300
1	0.0000	0.1	1.000000	0.791800	0.798900	0.710100	0.857100
2	0.0000	0.3	0.987600	0.781100	0.843100	0.623200	0.907300
3	0.0001	0.0	1.000000	0.802600	0.797900	0.744000	0.849400
4	0.0001	0.1	1.000000	0.785400	0.761000	0.753600	0.810800
5	0.0001	0.3	0.999500	0.789700	0.768500	0.753600	0.818500
6	0.0010	0.0	0.995400	0.772500	0.718600	0.801900	0.749000
7	0.0010	0.1	0.991300	0.813300	0.852900	0.700500	0.903500
8	0.0010	0.3	0.995900	0.802600	0.794900	0.748800	0.845600
9	0.0050	0.0	0.932800	0.766100	0.731100	0.748800	0.779900
10	0.0050	0.1	0.993600	0.721000	0.653400	0.792300	0.664100
11	0.0050	0.3	0.969600	0.731800	0.820300	0.507200	0.911200

Fig 5.4b: Val Accuracy Heatmap — CNN4



Without regularization, the model achieves a training accuracy of 1.0000 and validation accuracy of 0.8047 — the highest unregularized validation accuracy across all models tested. Batch normalization is clearly providing an implicit regularization effect, smoothing the loss landscape and allowing the network to converge to a more generalizable solution even without explicit penalties.

With regularization, as shown in Table 5.4, CNN4 produces the strongest and most consistent results of any architecture. The best configuration is L2=0.001, Dropout=0.1 (Val Acc: 0.8133, Precision: 0.8529, Recall: 0.7005, Specificity: 0.9035) — the highest validation accuracy achieved across all models in the entire report. The precision of 0.8529 is particularly impressive, meaning that when the model predicts Asian, it is correct 85% of the time. The heatmap (Fig 5.4b) shows that the sweet spot lies around low-to-moderate L2 with light dropout,

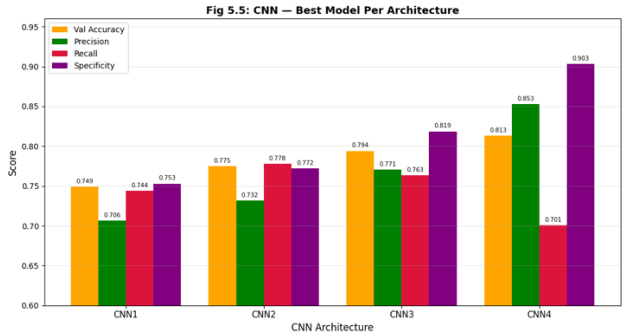
with performance degrading at stronger regularization values where batch normalization and explicit regularization begin to over-constrain the network together.

One tradeoff worth noting is that the best configuration's recall of 0.7005 is lower than its precision, meaning the model still misses some Asian faces. This precision-recall asymmetry reflects the mild class imbalance in the dataset and suggests the model is slightly conservative in predicting the minority class. Nevertheless, CNN4 represents a substantial improvement over all previous architectures, demonstrating that batch normalization combined with a deeper classifier head is the most effective design choice explored in this project.

5.5 CNN Summary

Table 5.5: CNN — Best Model Per Architecture

	Architecture	Key Feature	Best L2	Best Dropout	Train Acc	Val Acc	Precision	Recall	Specificity
0	CNN1	Single Conv Layer	0.0010	0.1	0.992600	0.748900	0.706400	0.744000	0.752900
1	CNN2	Two Conv Layers	0.0010	0.3	0.995400	0.774700	0.731800	0.777800	0.772200
2	CNN3	Three Conv Layers	0.0000	0.3	0.998600	0.794000	0.770700	0.763300	0.818500
3	CNN4	Three Conv + Batch Norm	0.0010	0.1	0.991300	0.813300	0.852900	0.700500	0.903500



Across all four CNN architectures, the best model is CNN4 with L2=0.001, Dropout=0.1, achieving a validation accuracy of 0.8133, precision of 0.8529, recall of 0.7005, and specificity of 0.9035. The progression from CNN1 to CNN4 tells a clear story: each architectural addition — a second convolutional layer, a third convolutional layer, and finally batch normalization with a deeper FC head — contributed meaningfully to improved generalization, confirming that spatial hierarchies are genuinely important for this task.

A key observation across all CNN architectures is that training accuracy remains near 1.0000 throughout, yet validation accuracy steadily improves from CNN1 to CNN4. This means the CNNs are not simply overfitting less — they are learning increasingly better representations. Batch normalization in CNN4 plays a dual role: it acts as an implicit regularizer while simultaneously stabilizing training, allowing the network to generalize without requiring aggressive explicit penalties.

Compared to logistic regression and fully connected networks, CNNs are the clear winner for this image classification task. The spatial inductive bias of convolutional filters — detecting local patterns regardless of position — is far better suited to facial image data than treating pixels as independent features.

6. Conclusion & Analytical Discussion

6.1 Overall Best Model & Test Performance

The best model identified across all experiments is CNN4 with L2=0.001 and Dropout=0.1. When evaluated on the held-out test set, it achieves a test accuracy of 0.7345, precision of 0.7079, recall of 0.6875, and specificity of 0.7722. The gap between validation accuracy (0.7704)* and test accuracy (0.7345) is modest, suggesting the model generalizes reasonably well beyond the validation set rather than having been over-tuned to it.

6.2 What Helped and Why

Several consistent patterns emerged across all three model families:

Feature representation was the single most important factor. For logistic regression, raw pixels with L2 regularization outperformed PCA-compressed features because L2 suppresses noise without discarding discriminative signal. For neural networks, switching to PCA input dramatically improved stability and generalization despite a smaller architecture. For CNNs, the spatial inductive bias of convolutional filters proved most naturally aligned with the image classification task, consistently outperforming fully connected alternatives regardless of architecture size.

Regularization played a critical but nuanced role. Across all models, the optimal regularization was moderate — too little led to overfitting while too much collapsed recall by biasing predictions toward the majority class. For CNNs, batch normalization in CNN4 provided implicit regularization that complemented explicit L2 and dropout penalties, making it the most stable and highest-performing architecture overall.

Architecture depth showed clear diminishing returns for fully connected networks but consistent gains for CNNs. Each additional convolutional layer in CNN1 through CNN4 produced meaningful improvements in validation accuracy, confirming that hierarchical spatial feature extraction is genuinely beneficial for facial image data. In contrast, adding layers to fully connected networks on raw pixels consistently worsened performance due to overfitting.

6.3 Bias-Variance Analysis

Throughout the experiments, the fundamental challenge was managing the bias-variance tradeoff given a moderately sized dataset of ~3,100 images:

Logistic regression suffered from high bias — its linear decision boundary is fundamentally insufficient for separating facial images with complex nonlinear structure. Even the best logistic model plateaued around 0.73 validation accuracy. Feature transformations like polynomial expansions attempted to increase model capacity but introduced excessive variance that regularization could not fully control.

Fully connected neural networks on raw pixels suffered primarily from high variance. The 12,288-dimensional input overwhelmed the networks' ability to generalize, and architecture choices had relatively little impact — the bottleneck was the input representation itself.

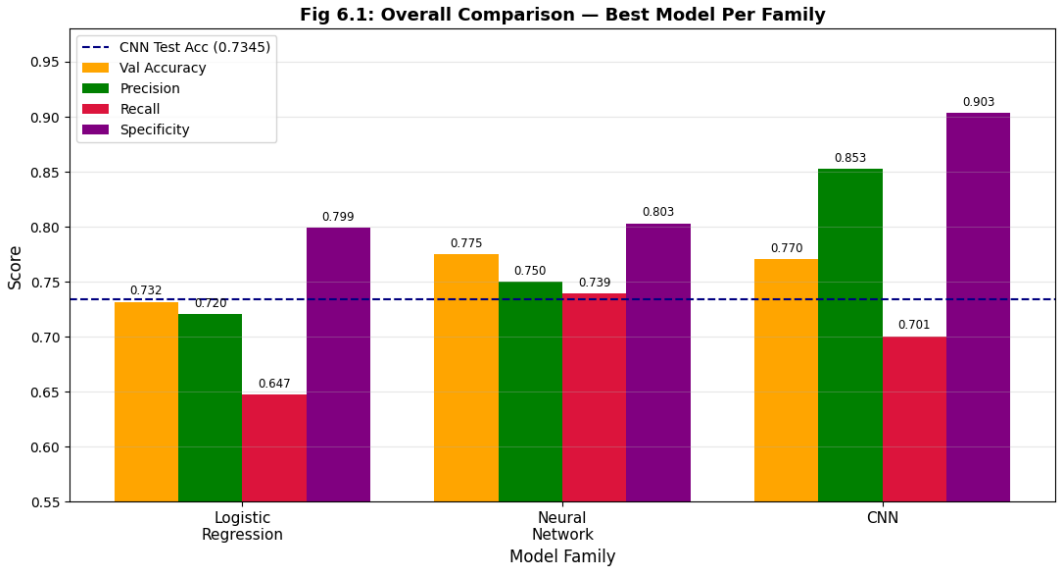
Switching to PCA input resolved this by reducing variance at the cost of some bias, resulting in the best fully connected model.

CNNs achieved the best bias-variance balance. Their shared convolutional weights dramatically reduce the effective number of free parameters relative to a fully connected network of similar depth, lowering variance without sacrificing the capacity to learn complex spatial patterns. Batch normalization in CNN4 further stabilized training and pushed the balance toward lower variance, producing the strongest overall generalization.

6.4 Limitations & Future Direction

Several limitations are worth acknowledging. First, the dataset size of ~3,100 images is relatively small for image classification, which likely prevented deeper or wider architectures from reaching their full potential. Second, all models were trained on 64×64 RGB images — higher resolution inputs might allow convolutional filters to capture finer facial details. Third, the mild class imbalance (44.5% vs 55.5%) contributed to the persistent precision-recall asymmetry observed across models, particularly the tendency to under-predict the Asian class at higher regularization strengths.

Future directions worth exploring include data augmentation (random flips, crops, color jitter) to artificially expand the training set and reduce overfitting, transfer learning from a large pretrained CNN such as ResNet or VGG which would provide far richer feature representations than training from scratch, and class-weighted loss functions to better handle the class imbalance.



*CNN's validation accuracy is different from the previous one because I retrain the model for the test set.

7. Reference

baig, Talha (2025), “Asian and Non-Asian People Datasets for Classification of Facial images”, Mendeley Data, V1, doi: 10.17632/6wcczyfdrb.1
https://data.mendeley.com/datasets/6wcczyfdrb/1